and if the sol particles migrate to the anode then they have negative charges(-). Thus from the direction of migration of sol particles we can determine the charge of the sol particles. Hence electrophoresis is used for detection of presence of charges on the sol particles.

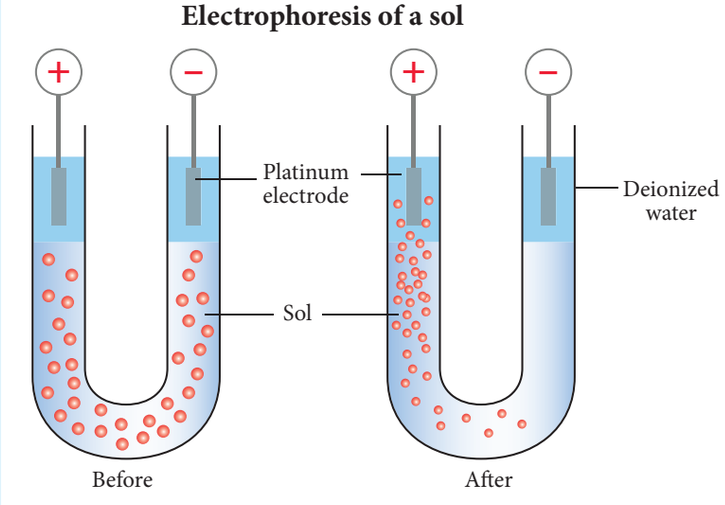


Figure 10.15 Electrophoresis

Few examples of charges of sols detected by electrophoresis are given below:

Positively charged colloids	Negatively charged colloids
Ferric hydroxide	Ag, Au & Pt
Aluminium hydroxide	Arsenic sulphide
Basic dyes	Clay
Haemoglobin	Starch

(iii) Electro osmosis

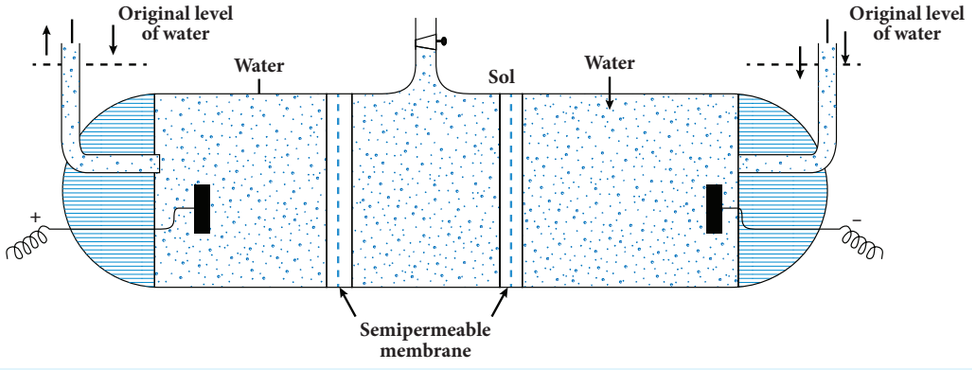


Figure 10.16 Electro osmosis

A sol is electrically neutral. Hence the medium carries an equal but opposite charge to that of dispersed particles. When sol particles are prevented from moving, under the influence of electric field the medium moves in a direction opposite to that of the sol particles. This movement of dispersion medium under the influence of electric potential is called electro osmosis.

XII U10-Surface Chemistry.indd 95 16-12-2022 19:40:33